

# HAIC 2026 Strategy and Experiment Control Report

Three-lane comparison, literature mapping, and submission-safe improvement workflow | 2026-09-22

## Executive conclusion

The current submission baseline is **ppo\_actor\_only**. It completed 6 of 8 held-out episodes (0.75 completion rate), with a 19,320 ms median finished lap and 22,420 ms P90 finished lap. It is the only lane currently backed by a packaged submission archive and CPU smoke result.

The **ppo\_cem** lane is not promoted: in the paired fast-budget held-out comparison it completed 0 of 10 episodes and reached mean progress 0.074530, below PPO-only at 0.077411. The **vision\_corridor\_teacher** lane remains a teacher and diagnostic reference, not a submission runtime.

## Three-lane comparison contract

| Lane                    | Role                   | Promotion evidence                                      |
|-------------------------|------------------------|---------------------------------------------------------|
| ppo_actor_only          | Submission candidate   | Strict held-out or official improvement                 |
| ppo_cem                 | Model-based candidate  | Same-split improvement after latency and package checks |
| vision_corridor_teacher | Teacher and diagnostic | Evidence only; never direct submission runtime          |

Comparison order: completion rate, median finished lap time, P90 finished lap time, unfinished progress, collisions and damage, then inference cost. Tune selects experiments; held-out and official evidence decide SOTA.

## Evidence and provenance

| Evidence           | Observation                                                    | Source                                                         |
|--------------------|----------------------------------------------------------------|----------------------------------------------------------------|
| PPO actor held-out | 6/8 complete; progress 0.788044; median 19320 ms; P90 22420 ms | artifacts/haic/final-ppo-actor-selection.json                  |
| PPO plus CEM       | 0/10 complete; mean progress 0.074530                          | artifacts/haic/task5-eval-fast-fullcap/summary.json            |
| Corridor teacher   | 3/3 complete; mean progress 0.996622                           | artifacts/haic/corridor-controller-candidate-v3/benchmark.json |
| Submission smoke   | Planner disabled; finite action; about 203 MB RSS              | artifacts/haic/final-ppo-actor-selection.json                  |

## Literature to experiment mapping

**PPO.** The model-free actor baseline is tested with fixed seeds, identical maps, decision cap, CPU budget, and package smoke. The paper supports the algorithm choice, not a claim about a particular HAIC checkpoint.

**DAGGER.** Learner-induced observations motivate train-only teacher corrections. Held-out maps must remain separate, and map geometry cannot enter submission inference.

**MBPO and short model rollouts.** Model error can harm policy quality. Any new CEM experiment must measure model error, planner latency, and the same held-out episodes before promotion.

**Domain randomization.** New local maps, obstacle layouts, and visual perturbations expand train and stress-test coverage. They do not replace official tracks or authorize hidden state at inference.

## Next experiment protocol

1. Freeze the current SOTA checkpoint and split manifest. 2. Select one hypothesis and one strategy lane. 3. Write to a new timestamped artifact directory. 4. Run train, tune, held-out, official diagnostics, package validation, and CPU smoke. 5. Sync RESULTS.md, compare all lanes, and update SOTA.md only when restriction and strict-improvement gates pass.

## References

1. Schulman et al., *Proximal Policy Optimization Algorithms*, 2017. <https://arxiv.org/abs/1707.06347>
2. Ross, Gordon, and Bagnell, *A Reduction of Imitation Learning and Structured Prediction to No-Regret Online Learning*, AISTATS 2011. <https://proceedings.mlr.press/v15/ross11a.html>
3. Janner et al., *When to Trust Your Model: Model-Based Policy Optimization*, NeurIPS 2019. <https://papers.nips.cc/paper/2019/hash/5faf461eff3099671ad63c6f3f094f7f-Abstract.html>
4. Tobin et al., *Domain Randomization for Transferring Deep Neural Networks from Simulation to the Real World*, 2017. <https://arxiv.org/abs/1703.06907>

This report records measured evidence and experiment design. It does not claim an improvement or successful external submission before those checks are completed.